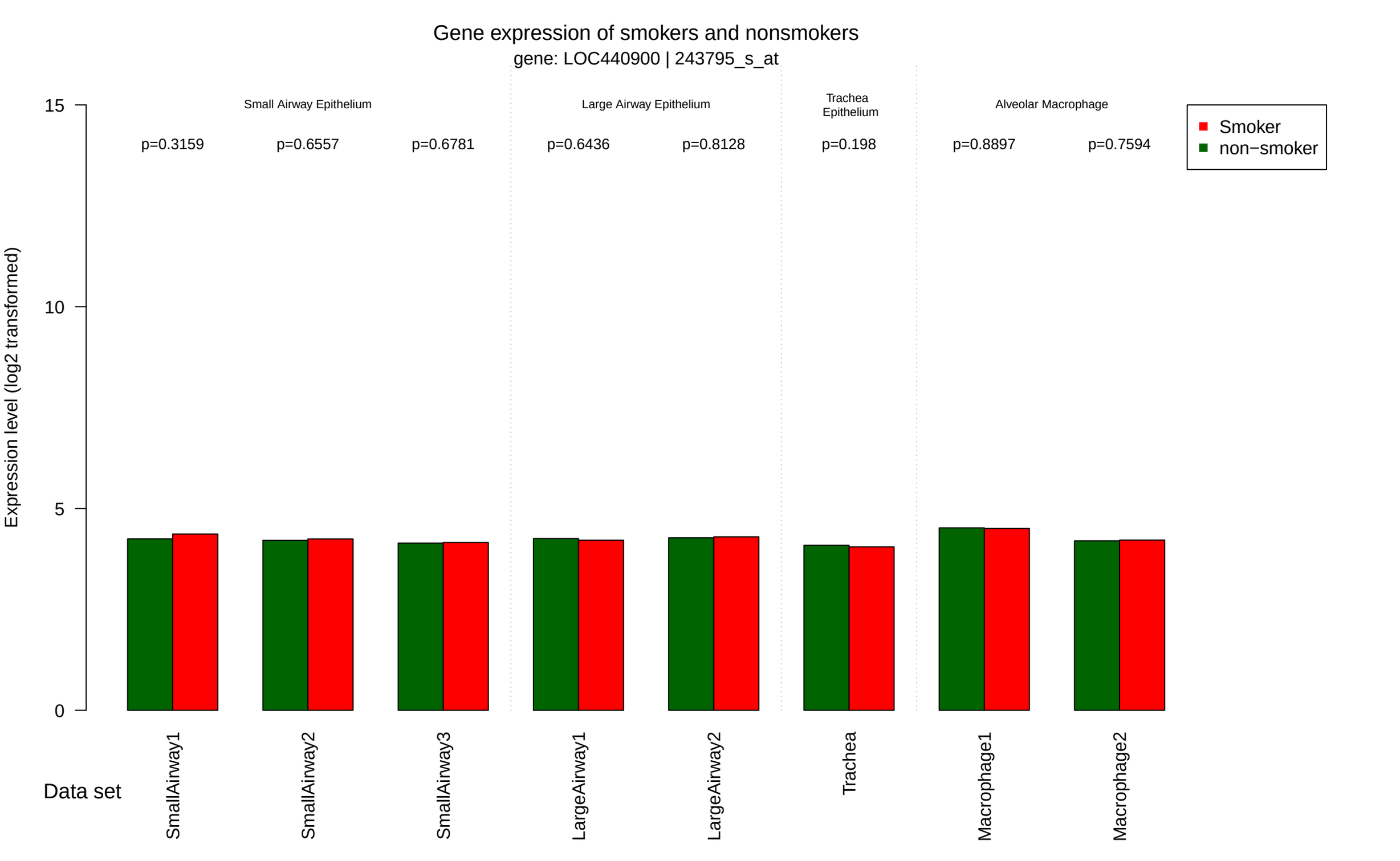
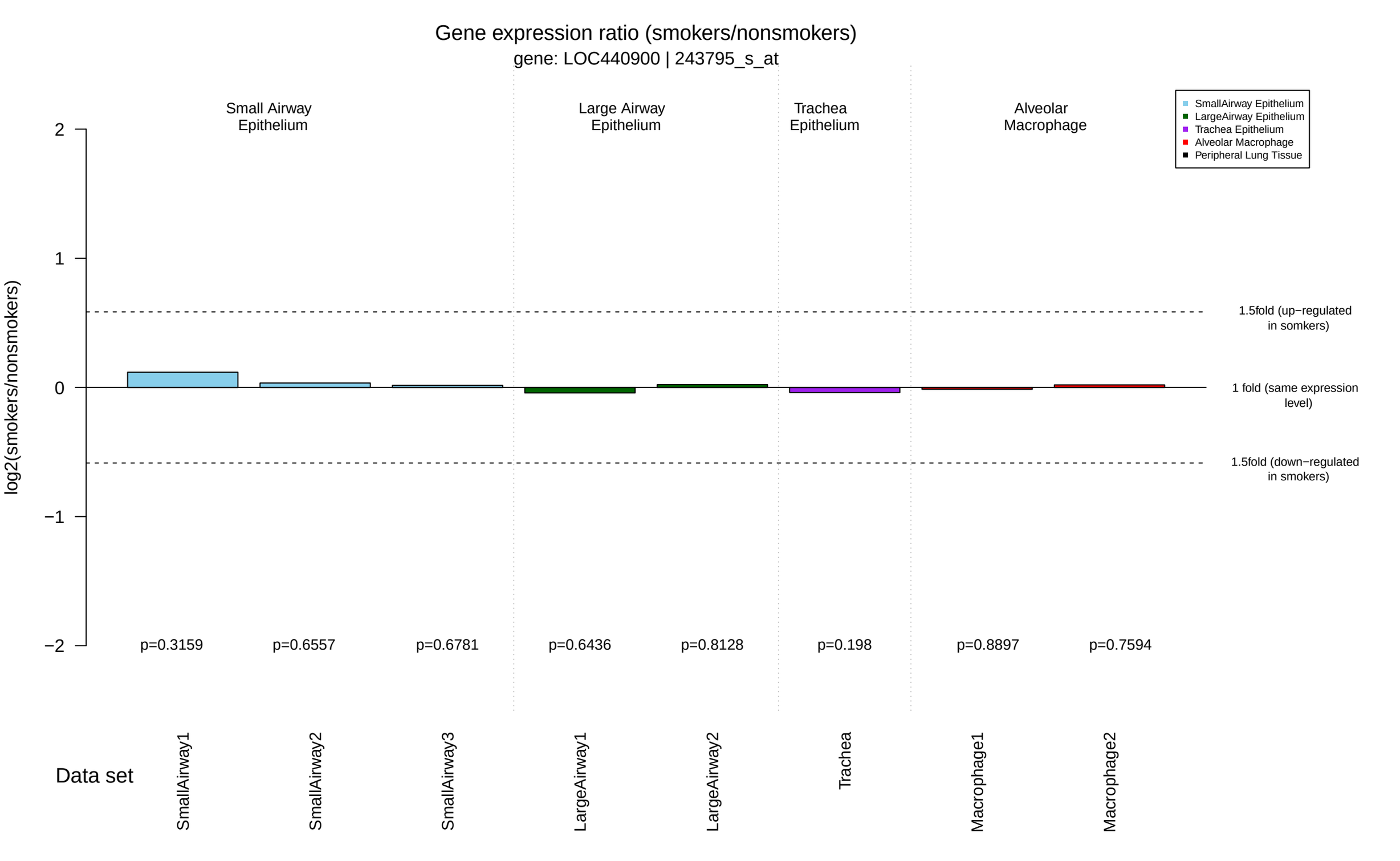
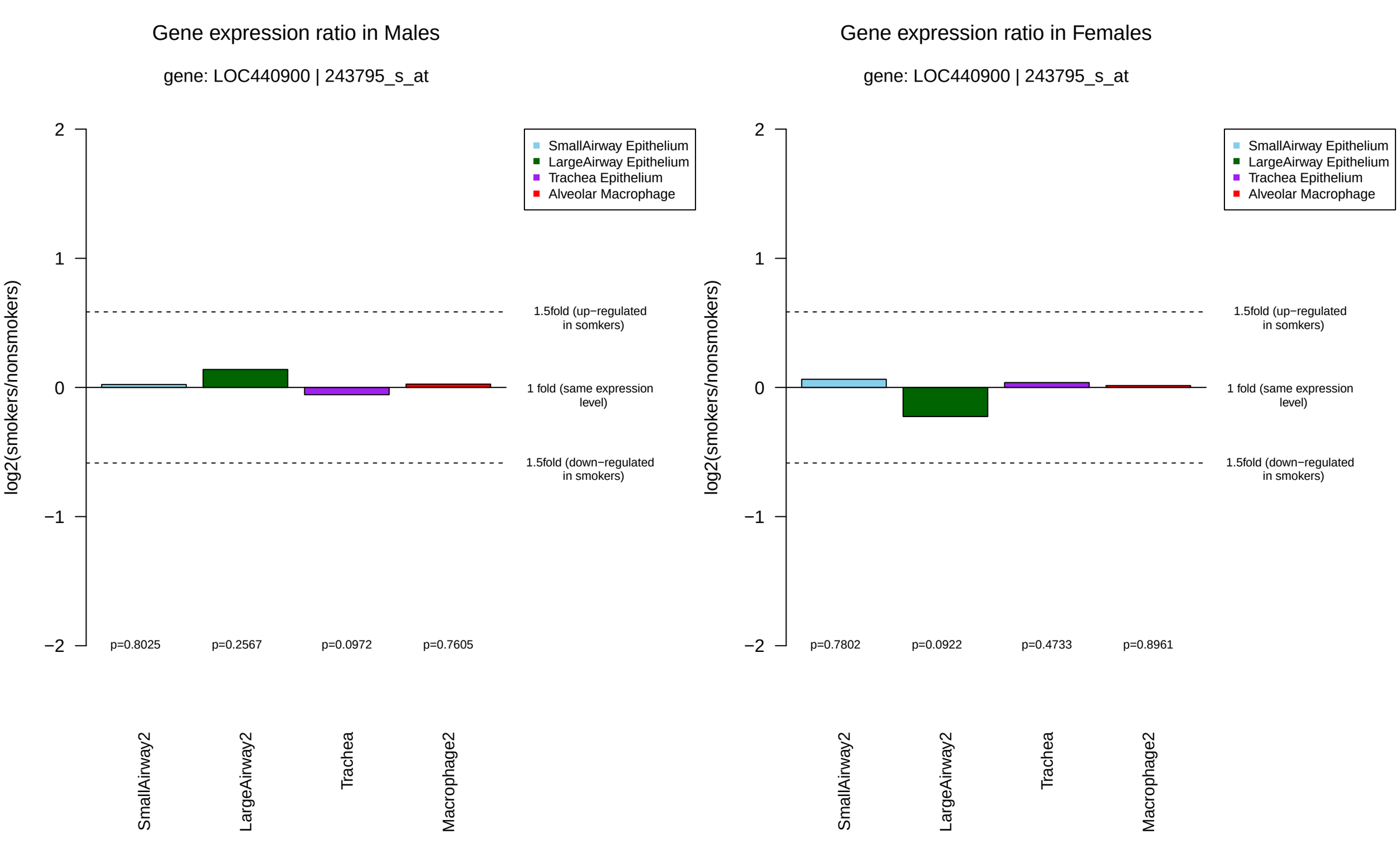
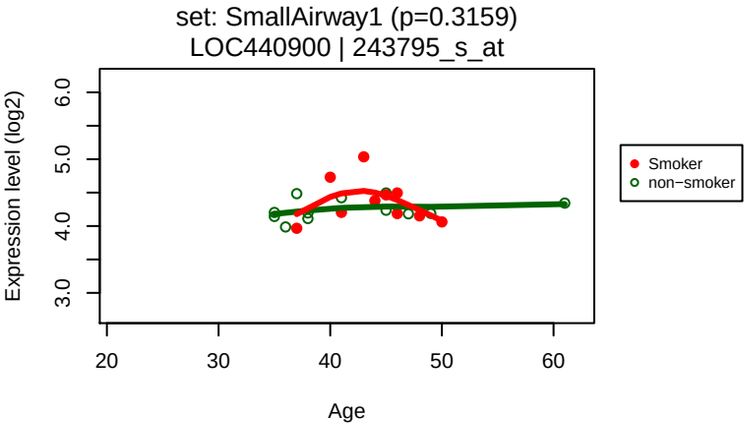
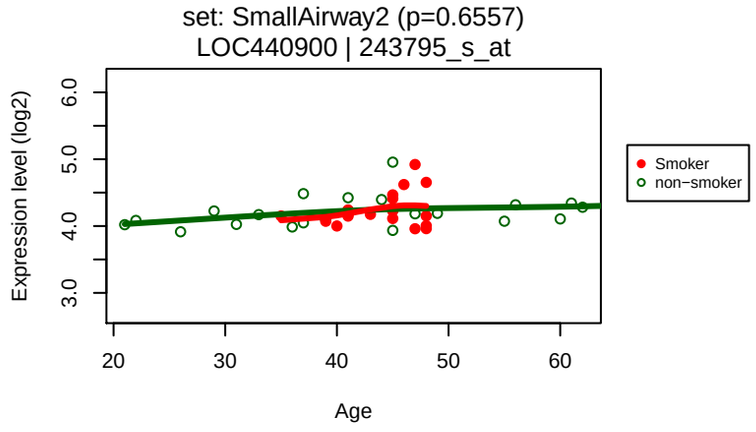
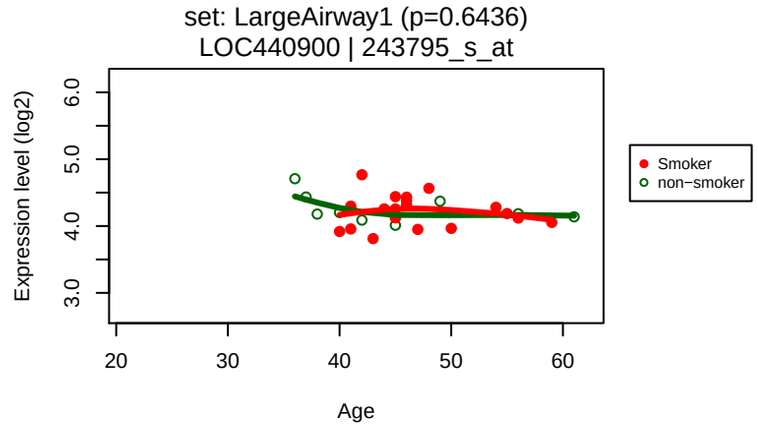

Gene expression of smokers and nonsmokers
gene: LOC440900 | 243795_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.3159
p=0.6557
p=0.6781
p=0.6436
p=0.8128
p=0.198
p=0.8897
p=0.7594
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

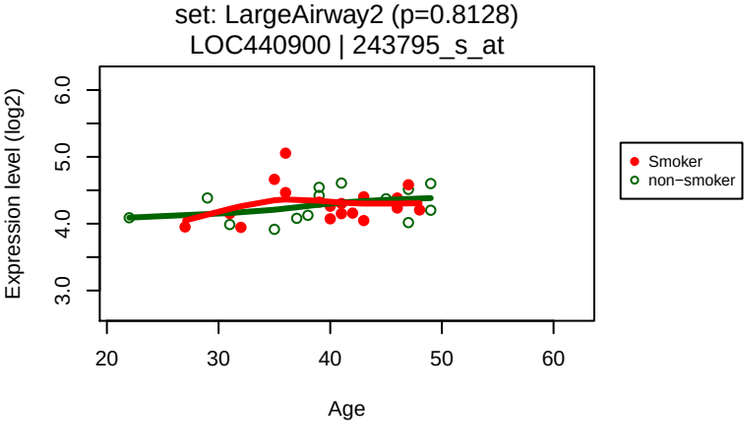

Gene expression ratio (smokers/nonsmokers)
gene: LOC440900 | 243795_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3159
p=0.6557
p=0.6781
p=0.6436
p=0.8128
p=0.198
p=0.8897
p=0.7594
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

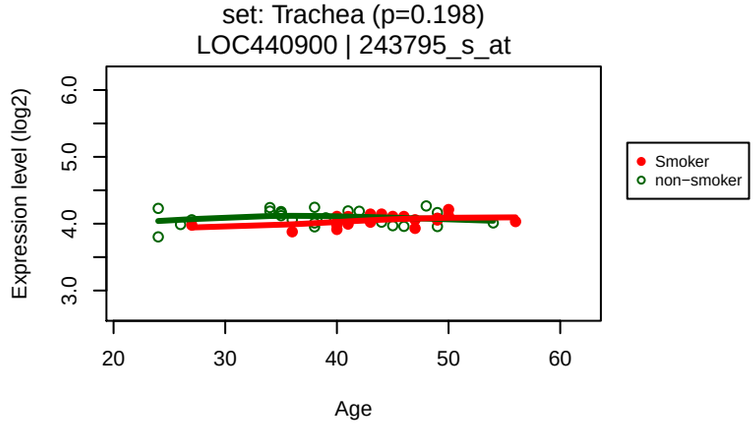

Gene expression ratio in Males
gene: LOC440900 | 243795_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8025
p=0.2567
p=0.0972
p=0.7605
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: LOC440900 | 243795_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.7802
p=0.0922
p=0.4733
p=0.8961
SmallAirway2
LargeAirway2
Trachea
Macrophage2

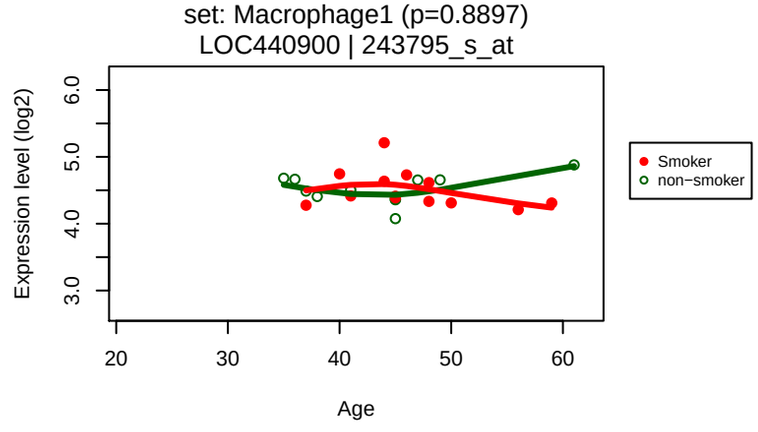

set: SmallAirway1 (p=0.3159)
LOC440900 | 243795_s_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

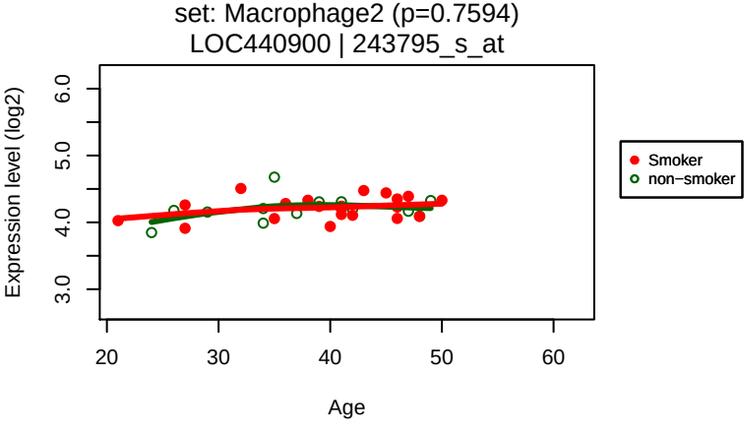

set: SmallAirway2 (p=0.6557)
LOC440900 | 243795_s_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker

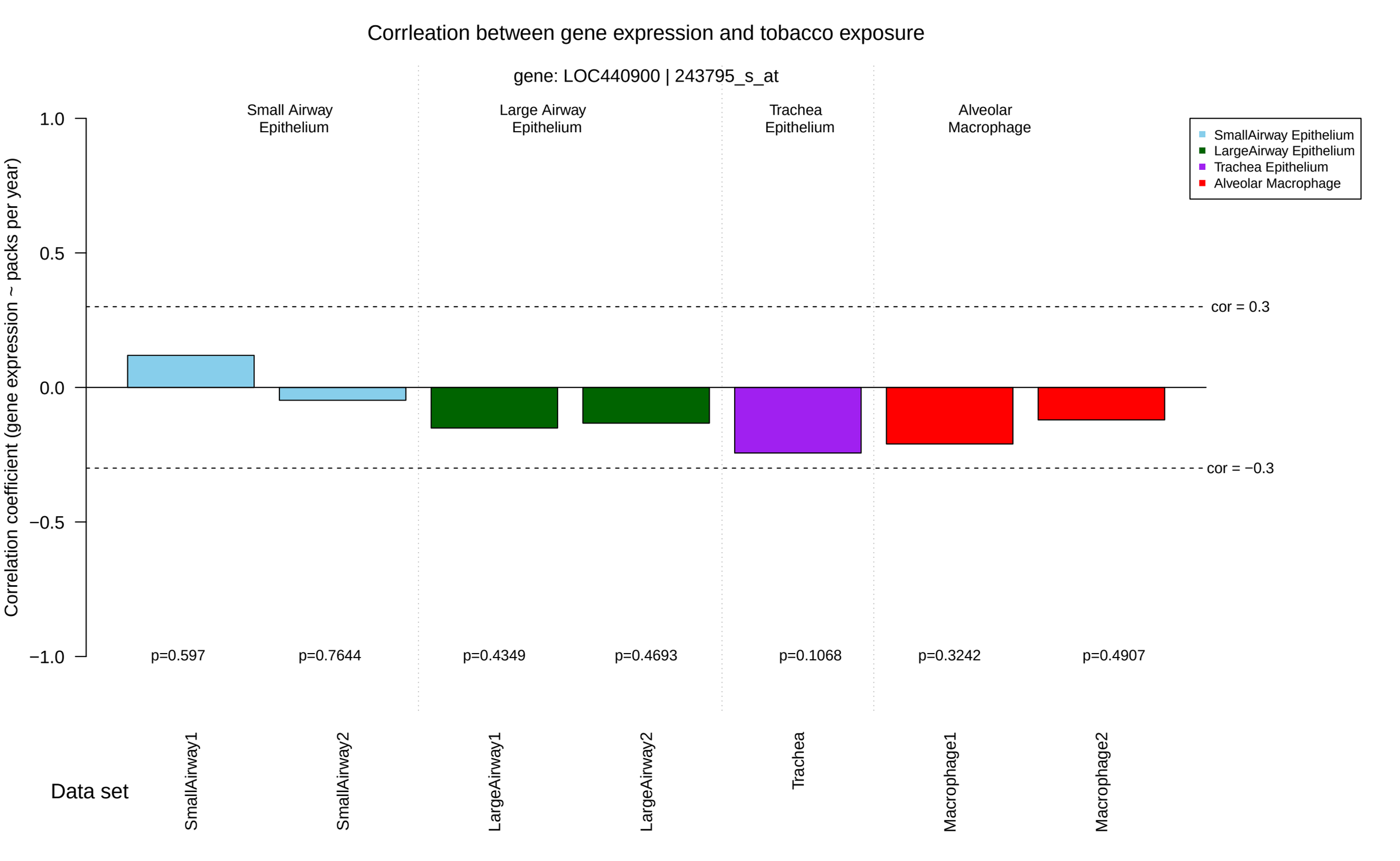

set: LargeAirway1 (p=0.6436)
LOC440900 | 243795_s_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8128)
LOC440900 | 243795_s_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Trachea (p=0.198)
LOC440900 | 243795_s_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8897)
LOC440900 | 243795_s_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7594)
LOC440900 | 243795_s_at
Expression level (log2)
3.0
4.0
5.0
6.0
20
30
40
50
60
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: LOC440900 | 243795_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.597
p=0.7644
p=0.4349
p=0.4693
p=0.1068
p=0.3242
p=0.4907
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2